PART A: SUPERVISED LEARNING PRACTICAL

Employee Attrition Prediction

Workplace Case

A company wants to understand why some employees leave while others stay. You are given a dataset containing:

- **Age**
- **Salary**
- **Years Employed**
- **Left Company (Yes/No)**

The variable **Left Company** represents the known outcome:

- **1 = Employee resigned**
- **0 = Employee stayed**

Because the outcome is already known, this is a **supervised learning problem**.

📎 Dataset for practical work: Download CSV file

STUDENT PRACTICAL QUESTIONS

Question 1

Understanding the Learning Problem

Identify which variables in the dataset are:

- input features
- target variable

Then explain why employee attrition prediction is classified as **supervised learning rather than unsupervised learning**.

Student Instruction

Your answer must explain:

- what makes a variable a feature,
- what makes a variable a label,

- why the presence of known outcomes matters in machine learning.

Question 2

Workplace Interpretation of Variables

Examine the dataset carefully and discuss which variable you believe may have the strongest influence on whether an employee leaves the company.

Choose one of the following and justify your answer:

- age
- salary
- years employed

Student Instruction

Your justification must be based on workplace reasoning, not only observation.

For example:

- Why might low salary increase resignation risk?
- Why might short employment duration matter?

Question 3

Predictive Modeling Decision

A human resources department wants to predict whether a newly recruited employee is likely to leave within two years.

Would this problem require:

- regression
- classification
- clustering

Choose one and justify your answer professionally.

Student Instruction

Your answer must explain:

- the nature of the target variable,
- why the selected method fits the problem,
- why the other methods are less suitable.

Question 4

Interpreting a New Employee Case

Suppose a new employee has the following characteristics:

- Age: 28 years
- Salary: 3500
- Years employed: 2 years

Based on the patterns in the dataset, discuss whether this employee may have a high or low probability of leaving the company.

Student Instruction

You are not required to calculate exact probability.

You must explain:

- which variables influence your judgment,
- what assumptions you are making,
- why prediction in machine learning is probabilistic rather than certain.

Question 5

Limits of Supervised Learning in Human Resource Decisions

Even if a machine learning model predicts attrition correctly during training, explain one reason why it may fail when used in a real company.

Student Instruction

Use one practical limitation such as:

- missing variables,
- changing employee behaviour,

- biased historical data,
- economic changes.

Your answer must show critical understanding of real-world deployment.

Advanced Reflection Question 🚀

Ethical Thinking

Should a company rely entirely on machine learning to decide which employees are likely to resign?

Explain one risk of using prediction without human judgment.

Submission Requirement

Students must submit:

✅written answers
✅ notebook analysis
✅ interpretation of findings

Not accepted:

❌ short one-line answers
❌ definitions copied from internet